Cephalic phase of colonic pressure response to food.

A cephalic phase of colonic pressure response to food was sought in five normal subjects (mean age (22.6) years, 22-24), studied on six separate occasions by recording intraluminal pressures in the unprepared sigmoid colon. Gastric acid secretion was measured simultaneously by continuous aspiration through a nasogastric tube. After a 60 minute basal period, one of five 30 minute food related cephalic stimuli, or a control stimulus was given in random order; records were continued for a further 120 minutes. The cephalic stimuli were: food discussion, sight and smell of food without taste, smell of food without sight or taste, sight of food without smell or taste, and modified sham feeding; the control stimulus was a discussion of neutral topics. Colonic pressures were expressed as study segment activity index (area under curve, mm Hg.min) derived by fully automated computer analysis. Gastric acid output was expressed as mmol/30 min. Food discussion significantly ($p < 0.02$, Wilcoxon's rank sum test) increased colonic pressure activity compared with control or basal activity. Smell of food without sight or taste also significantly ($p < 0.03$) increased the colonic pressure activity compared with control and basal periods. Sham feeding and sight and smell of food without taste significantly ($p < 0.02$ and $p < 0.03$) increased colonic pressures compared with control but not basal activity. The increase in colonic activity after sight of food without smell or taste was not significantly different from control or basal activity ($p = 0.44$ and $p = 0.34$). Food discussion was the strongest colonic stimulus tested. Food discussion and sham feeding significantly ($p<0.02$) stimulated gastric acid output above control and basal values. Sight and smell of food without taste significantly ($p<0.02$) increased acid output above basal. Smell of food without sight or taste and sight of food without smell or taste did not significantly ($p=0.06$, $p=0.34$) increase acid output. In contrast with the effect on colonic pressures, sham feeding was the best stimulus of acid output. Increased colonic pressure activity after food discussion correlated significantly ($r=0.45$, $p<0.02$) with gastric acid output. There was no correlation ($r=-0.1$, $p>0.5$) between colonic pressure activity and gastric acid output in the control study. These data show that there is a cephalic phase of the colonic response to food.